

Exhibit 38

DRAM COMPONENTS

DDR5 DRAM



16Gb
Component Density

2400MHz
Speed

-40C to +105C
Operating Temp

DDR5 SDRAM

16Gb, 1.1 VOLTS, -40C to +105C, 2400MHz

Where to buy

Complete parts catalog

Component Density

16Gb

24Gb

32Gb

I/O Voltage

1.1 VOLTS

Operating Temp

-40C to +105C

-40C to +95C

0C to +95C

Speed

2400MHz

2600MHz

2800MHz

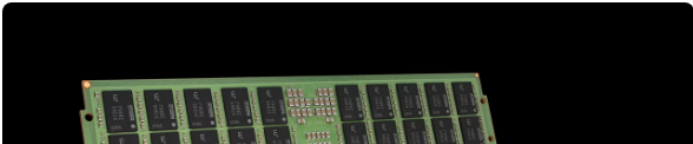
3200MHz

3600MHz

Matching Products (2)

Export Matching Products 

PART NUMBER	DOWNLOAD	WHERE TO BUY	SPEED	PART STATUS CODE	TAPE & REEL QTY	COMPONENT DENSITY	I/O VOLTAGE	NUM
MT60B2G8HS-48B AAT:A		Where to buy	2400MHz	Obsolete	3000	16Gb	1.1 VOLTS	1
MT60B1G16HT-48B AAT:A		Where to buy	2400MHz	Obsolete	2000	16Gb	1.1 VOLTS	1



Optimal memory performance and capacity for Intel® Xeon® 6 processors

micron

Products

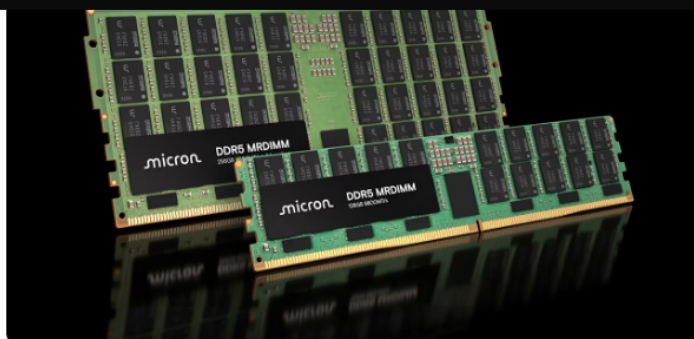
Market & Industries

Partners

Sales & Support

About

Search



Capacity for Intel® Xeon® 6 processors

Micron MRDIMM provides the highest-bandwidth, lowest-latency and highest-capacity main memory for accelerating memory-intensive workloads like AI and High-Performance Computing environments using Intel Xeon 6 processors.

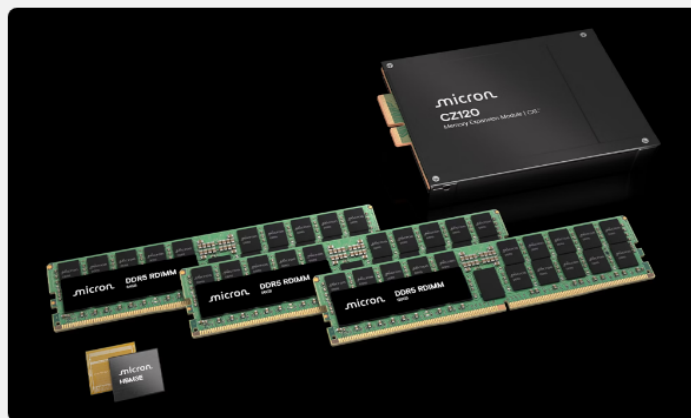
Available from 32GB up to 256GB, our high-capacity MRDIMMs are ready to be deployed in data center environments. Micron's high-performance MRDIMM solution delivers up to a 39% improvement in bandwidth with energy efficiency.

[LEARN MORE ABOUT MRDIMM](#)
[MRDIMM PRODUCT BRIEF](#)

Industry's first scalable memory

Micron delivers higher performance DRAM that:

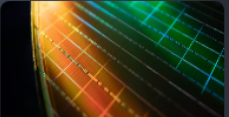
- Scaled overall bandwidth with DDR5 RDIMM speeds up to 9200 MT/s and MRDIMM up to 8800 MT/s
- Delivers up to a 2x improvement in memory bandwidth compared to DDR4 SDRAM at 3200 MT/s
- Brings new and increased capacities with 32Gb monolithic die based 128GB RDIMMs

[CONTACT SALES TEAM](#)


Fueling AI advances with Micron 96GB and 128GB high-capacity RDIMMs

Micron's world-class high-capacity RDIMM modules are fueling advances in AI data centers around the world. Powered by Micron's industry-leading 1β (1-beta) technology, the 128GB DDR5 RDIMM memory delivers more than 45% improved bit density,⁴ up to 22% improved energy efficiency⁵ and up to 16% lower latency⁴ over competitive 3DS through-silicon via (TSV) products.

[READ THE PRESS RELEASE](#)
[96GB TECH BRIEF](#)
[128GB PRODUCT BRIEF](#)

Pioneering memory technology advancements for future compute needs



compute needs

Micron today announced it is the first in the industry to sample its 1γ (1-gamma) DRAM node in DDR5 memory to select data center customers, client customers and ecosystem enablers.

[READ THE PRESS RELEASE](#)

Micron DDR5 SDRAM: Scalable memory for growing processor core counts

Fueling performance for modern workloads by delivering the memory bandwidth needed to turn large data sets into insight — quickly and efficiently.

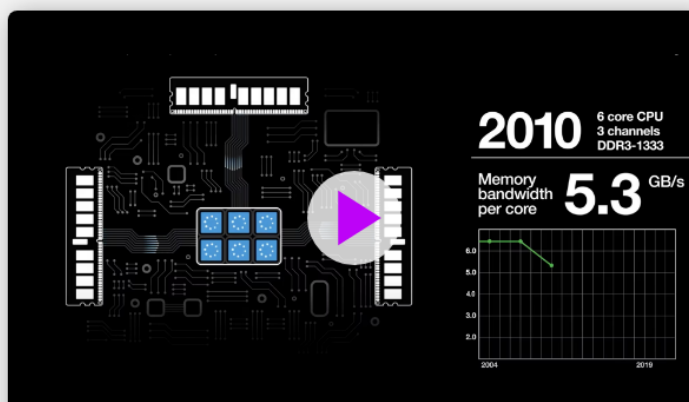
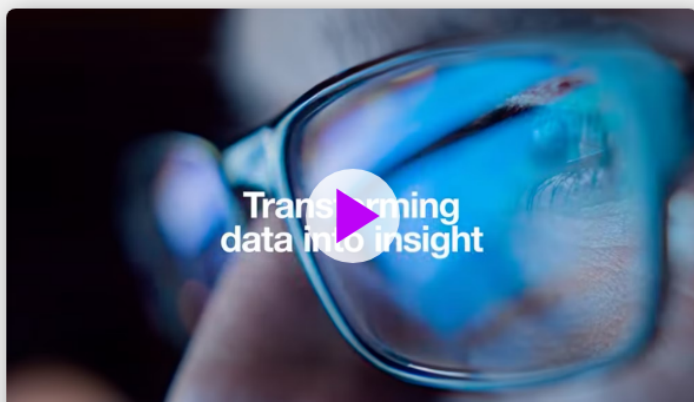
[DDR5 DRAM INFOGRAPHIC](#)

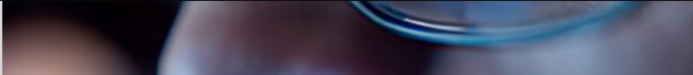
Technology Enablement Program for DDR5

The Technology Enablement Program (TEP) for DDR5 offers a path into Micron to gain early access to technical information and support, electrical and thermal models, as well as memory products to aid in the design, development and introduction of next-generation computing platforms.

[TEP PROGRAM DETAILS](#)

The advantages of DDR5





Micron DDR5: Offering more than 2X the effective bandwidth

DDR5 offers up to 2X the effective bandwidth compared to its predecessor DDR4, helping relieve the bandwidth-per-core crunch.

[Watch DDR5 2X bandwidth video →](#)



Micron DDR5: Next-generation memory transforming data into insight

Watch how Micron DDR5 addresses the memory bandwidth-per-CPU challenge and transforms available data into insight.

[Watch DDR5 next-gen video →](#)

Micron DDR5: The memory technology for next-generation Intel server platforms

Intel has announced the latest generation of servers featuring 4th Gen Intel® Xeon® Scalable processors. Micron DDR5 runs SPECjbb 48% faster on the Dell PowerEdge 760 platform compared to previous generations.³

[READ MICRON DDR5 ARTICLE](#)

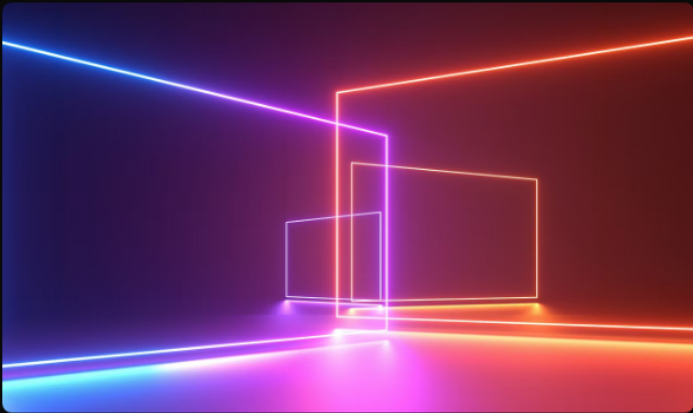


Advantages of migrating to DDR5

DDR5 is the next evolution in DRAM, bringing a robust list of new features geared to increase reliability, availability and serviceability (RAS); reduce power; and dramatically improve performance. See below for a list of some of the key feature differences between DDR4 and DDR5.

FEATURE/OPTION	DDR4	DDR5	DDR5 ADVANTAGE
Data rates	600-3200MT/s	4800-8800MT/s	Increases performance and bandwidth
VDD/VDDQ/VPP	1.2/1.2/2.5	1.1/1.1/1.8	Lowers power
Internal VREF	VREFDQ	VREFDQ, VREFCA, VREFCS	Improves voltage margins, reduces BOM costs
Device densities	2Gb-16Gb	16Gb, 24Gb, 32Gb	Enables larger monolithic devices
Prefetch	8n	16n	Keeps the internal core clock low
DQ receiver equalization	CTLE	DFE	Improves opening of the received DQ data eyes inside the DRAM
Duty cycle adjustment (DCA)	None	DQ and DQS	Improves signaling on the transmitted DQ/DQS pins
Internal DQS delay monitoring	None	DQS interval oscillator	Increases robustness against environmental changes
On-die ECC	None	128b+8b SEC, error check and scrub	Strengthens on-chip RAS

On-die ECC	None	128b+8b SEC, error check and scrub	Strengthens on-chip RAS
CRC	Write	Read/Write	Strengthens system RAS by protecting read data
Bank groups (BG)/banks	4 BG x 4 banks (x4/x8) 2 BG x 4 banks (x16)	8 BG x 4 banks (16-64Gb x4/x8) 4 BG x 4 banks (16-64Gb x16)	Improves bandwidth/performance
Command/address interface	ODT, CKE, ACT, RAS, CAS, WE, A<X:O>	CA<13:O>	Dramatically reduces the CA pin count
ODT	DQ, DQS, DM/DBI	DQ, DQS, DM, CA bus	Improves signal integrity, reduces BOM costs
Burst length	BL8 (and BC4)	BL16 (and BC8 OTF)	Allows 64B cache line fetch with only 1 DIMM subchannel.
MIR ("mirror" pin)	None	Yes	Improves DIMM signaling
Bus inversion	Data bus inversion (DBI)	Command/address inversion (CAI)	Reduces VDDQ noise
CA training, CS training	None	CA training, CS training	Improves timing margin on CA and CS pins
Write leveling training modes	Yes	Improved	Compensates for unmatched DQ-DQS path
Read training patterns	Possible with the MPR	Dedicated MRs for serial (user defined), clock and LFSR -generated training patterns	Makes read timing margin more robust
Mode registers	7 x 17 bits	Up to 256 x 8 bits (LPDDR type read/write)	Provides room to expand
PRECHARGE commands	All bank and per bank	All bank, per bank, and same bank	PREsb enables precharging a bank in each BG
REFRESH commands	All bank	All bank and same bank	REFsb enables refreshing a bank in each BG
Loopback mode	None	Yes	Enables testing of the DQ and DQS signaling



Micron's DDR5 ecosystem partners

Micron works closely with selected ecosystem partners to provide early access to next-generation storage and memory protocols. Learn how:

1. Micron partners with AMD to deliver next-generation data center performance.
2. Micron partners with Intel to deliver next-generation server platforms.

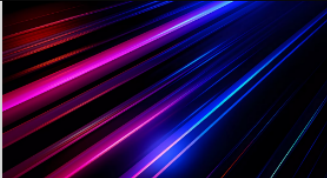
[ECOSYSTEM PARTNERS](#)

[AMD COLLABORATION](#)

[INTEL COLLABORATION](#)

Featured resources





Micron DDR5 delivers next-gen performance and reliability for 4th Gen Intel® Xeon® scalable processor family
[Read press release](#) →



Micron DDR5 memory now available for 4th Gen AMD EPYC processors
[Read press release](#) →



Boost HPC workloads with Micron DDR5 and 4th Gen AMD EPYC processors
[Read HPC article](#) →



Podcast: Micron DDR5 for the data center
[Listen to DDR5 podcast](#) →

Frequently asked questions

▣ What is DDR5 RAM?
DDR5 RAM is fifth-generation double data rate (DDR) random-access memory (RAM) technology. RAM is the general term people use to describe system memory, but all DDR5 memory is based on dynamic random access memory (DRAM) technology. DRAM stores data in memory cells made up of a capacitor and a transistor and is designed for fast data access while a system is running. “DDR” refers to the ability to transfer data on both the rising and falling edges of the clock signal, increasing data throughput.
+ Can you use DDR4 memory on a DDR5 motherboard?
+ How is DDR5 memory speed measured: MHz or MT/s?
+ What is the difference between SDRAM vs. DRAM vs. RAM?
+ What are the benefits of DDR4 vs. DDR5 RAM?
+ Is DDR5 server memory better than DDR4 in data centers?
+ What are the energy-saving benefits of DDR5 server RAM?
+ Why buy Micron DDR5 server RAM instead of competitive products?

1. STREAM benchmark testing: Single socket 3rd Gen AMD EPYC CPU 7763 (64 cores) with Micron DDR4 3200 MHz system is capable of 189 GB/sec; single socket 4th Gen AMD EPYC CPU 9654 (96 cores) with Micron DDR5 4800 MHz system is capable of 378 GB/sec; this result is based on testing at Micron Austin Labs.

2. Currently sampling to ecosystem partners.

3. <https://www.dell.com/en-us/blog/leading-sap-hana-performance-by-dell-poweredge-r760-servers/>

4. Based on competitive data sheets and JEDEC specifications.

5. Micron 5,600 MT/s 128GB DDR5 RDIMM memory compared to SK Hynix's 5,600 MT/s 3DS TSV product yields 22.2% better power efficiency.



Design tools
Access our design tools and resources to test, build, and maintain high-performance solutions to

Search for, filter and download data sheets
Get in-depth information about product features, specifications,

Order a Micron sample
Your online source for placing and tracking orders for Micron memory samples.

Downloads and technical documentation
Resources to help you design, test, build and optimize your innovative

 Can you use DDR4 memory on a DDR5 motherboard?
 How is DDR5 memory speed measured: MHz or MT/s?
 What is the difference between SDRAM vs. DRAM vs. RAM?
 What are the benefits of DDR4 vs. DDR5 RAM?
 Is DDR5 server memory better than DDR4 in data centers?
 What are the energy-saving benefits of DDR5 server RAM?
 Why buy Micron DDR5 server RAM instead of competitive products?

1. STREAM benchmark testing: Single socket 3rd Gen AMD EPYC CPU 7763 (64 cores) with Micron DDR4 3200 MHz system is capable of 189 GB/sec; single socket 4th Gen AMD EPYC CPU 9654 (96 cores) with Micron DDR5 4800 MHz system is capable of 378 GB/sec; this result is based on testing at Micron Austin Labs.
2. Currently sampling to ecosystem partners.
3. <https://www.dell.com/en-us/blog/loading-sap-hana-performance-by-dell-poweredge-r760-servers/>
4. Based on competitive data sheets and JEDEC specifications.
5. Micron 5,600 MT/s 128GB DDR5 RDIMM memory compared to SK Hynix's 5,600 MT/s 3DS TSV product yields 22.2% better power efficiency.

Design tools

Access our design tools and resources to test, build, and maintain high-performance solutions to support your design process.

[Visit design tools](#) →

Search for, filter and download data sheets

Get in-depth information about product features, specifications, functionality, and more.

[Search for data sheets](#) →

Order a Micron sample

Your online source for placing and tracking orders for Micron memory samples.

[Order samples](#) →

Downloads and technical documentation

Resources to help you design, test, build and optimize your innovative designs.

[Downloads and tech docs](#) →

Products

- Memory
- Storage
- Multichip packages
- Technology leadership
- Design tools

Markets & Industries

- AI
- Automotive
- Client PC
- Consumer
- Data center and servers
- Customer and partner storage success
- Industrial IoT
- Mobile
- Network infrastructure

Partners

- Partners overview
- Partner networks
- Technology enablement programs
- Become a partner
- Partner login
- Sales network
- Suppliers
- ODM Partners: AVL status by platform

Sales & Support

- Sales and support overview
- Customer support
- Sales
- Downloads and technical documentation

About

- About overview
- Company
- Corporate governance
- Micron blog
- Events
- Micron glossary
- Awards and recognition
- U.S. expansion

Investor Relations

- IR home
- Events and presentations
- Latest news
- Quarterly financial results